

MODULE-5

Sensors in Analytical Techniques

Electrode System

Electrochemistry: It is a branch of chemistry which deals with the study of transformation of chemical energy into electrical energy and vice versa”.

Electrochemical cells: “An electrochemical cell is a device which converts chemical energy into electrical energy or electrical energy into chemical energy”.

Types of Electrodes:

Metal-metal ion electrode: This type of electrode consists of a metal in contact with a solution of its own ions.

Egs: 1. Zinc rod in a solution of zinc sulphate.

2. Copper rod in a solution of copper sulphate.

Metal-metal salt ion electrode: Here a metal is dipped in its insoluble salt that is in contact with a solution containing the anion of the salt

Egs 1. Calomel electrode $\text{Hg} / \text{Hg}_2\text{Cl}_2(\text{s}) / \text{Cl}^-$

2. Silver-Silver chloride electrode $\text{Ag} / \text{AgCl}(\text{s}) / \text{Cl}^-$

Gas electrode: Here a gas is in contact with an inert metal that is dipped in an ionic solution of the gas molecule.

Egs: SHE, $\text{Pt} / \text{H}_2(1\text{atm}) / \text{H}^+(1\text{M})$

Ion-selective electrode: Possesses the ability to respond only to certain specific ions, thereby developing potential w.r.t that species only in a mixture and ignoring the other ions totally. In other words, the potential developed by an ion-selective electrode depends only on the concentration of species or ions of interest. For egs. Glass electrode which responds to hydrogen ions ($\text{Ag}/\text{AgCl}/\text{HCl}/\text{Glass}$).

Reference electrodes: Reference electrodes are the electrode with reference to those, the electrode potential of any electrode can be measured”.

Reference electrodes:

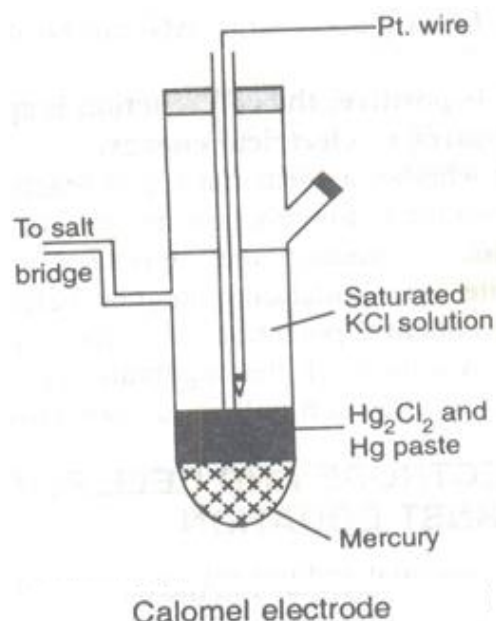
“Reference electrodes are the electrode with reference to those, the electrode potential of any electrode can be measured”.

There are two types of Reference electrodes

- 1) Primary reference electrode **EX;** - SHE
- 2) Secondary reference electrode **EX;** - Calomel Electrode, Ag-AgCl Electrode etc

Construction, working and applications of Calomel electrode.

Calomel electrode is a metal-metal salt ion electrode. It consists of mercury, mercurous Chloride and a solution of KCl. Mercury is placed at the bottom of a glass tube.

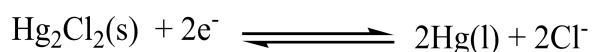


Calomel electrode consists of glass tube at the bottom of which mercury is placed. Above mercury a layer of mercury and mercurous chloride (called calomel) is placed. 3/4th of container is filled with saturated KCl solution. Pt wire is used for external electrical contact. Electrode potential of the cell depends on the concentration of KCl used. And

Calomel electrode can be represented as, **Hg|Hg₂Cl₂|KCl**

The calomel electrode can act as anode or cathode depending on the nature of the other electrode of the cell.

The net cell reversible electrode reaction is,



The Nernst equation is given by,

$$E = E^0 - \frac{2.303RT}{2F} \log[\text{Cl}^-]^2$$

$$E = E^0 - \frac{2.303RT}{F} \log[\text{Cl}^-]$$

$$E = E^0 - 0.0591 \log[\text{Cl}^-] \text{ at } 298 \text{ K}$$

The electrode potential of calomel electrode is depending upon the concentration of KCl used, $0.1N\ KCl \rightarrow 0.334V$, $1N\ KCl \rightarrow 0.281V$, $Sat.KCl \rightarrow 0.2422V$.

Advantages:-

1. Simple to Construct.
2. Cell potential does not vary with temperature.
3. The cell potential is reproducible and constant over a long period.

Applications:

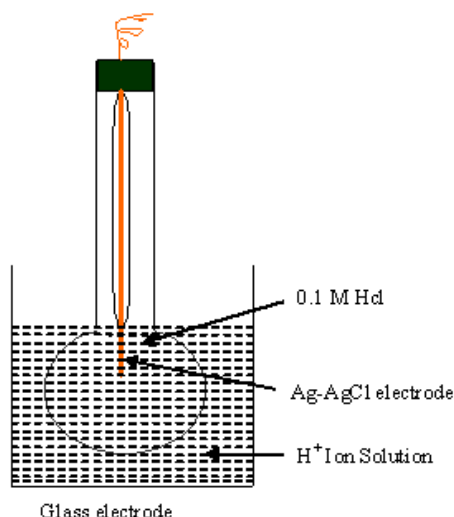
1. It is used as secondary reference electrode in the measurement of single electrode.
2. It is used as reference electrode in all potentiometer determinations.
- 3) The SCE is used in pH measurement

Ion-selective electrode:

“Ion selective electrode is one which selectively responds to specific ions in a mixture and the potential developed at the electrode is a function of the concentration of those ions in the solution”.

Uses:- To determine the concentration of a number of cations and anions such as H^+ , Li^+ , Na^+ , K^+ , Pb^{2+} , Cu^{2+} , Mg^{2+} , CN^- , NO_3^- , F^- etc.

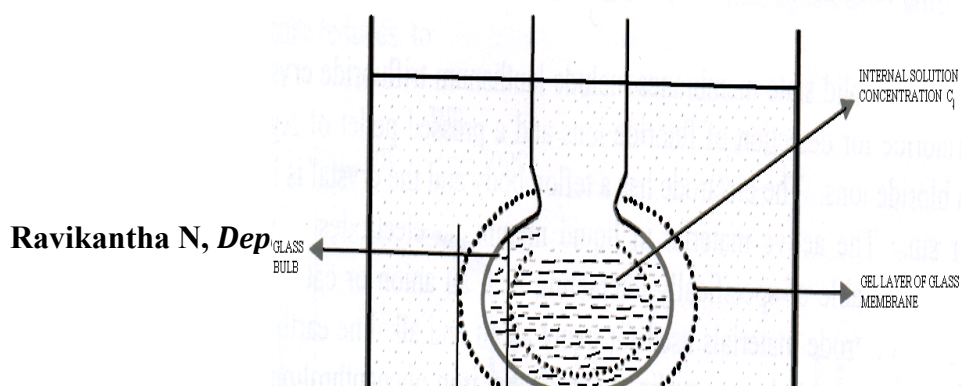
Construction, working and applications of Glass electrode.



A glass electrode is an ion selective electrode where potential depends upon the pH of the medium. The glass electrode consists of a glass bulb made up of special type of glass ($22\%Na_2O$, $6\%CaO$, and $72\%SiO_2$) which has low melting point and relatively high electrical conductivity. The glass bulb is filled with a solution of constant pH ($0.1MHCl$) and insert with a $Ag-AgCl$ electrode, which is the Internal reference electrode and also serves for the external electrical contact. The electrode dipped in a solution containing H^+ ions. The electrode representation is,

$Ag/AgCl / HCl (0.1M) / Glass$

Working - A thin walled glass bulb containing an acid is immersed in another solution a potential is developed across the glass membrane. The potential difference, E_b at the interface is referred to as boundary potential is the difference in potential (E_1-E_2) developed across the layer of the glass membrane b/w the two liquids.



The potential of glass electrode E_G is given by

$$E_G = E_1 - E_2 \quad \dots\dots\dots 1 \quad \text{Where, } E_1 \& E_2 \text{ are the electrode potential of outer \& inner membrane}$$

$$E_G = [E^\circ + 0.0591 \log [C_2]] - [E^\circ + 0.0591 \log [C_1]] \quad \dots\dots\dots 2 \quad \text{Where } C_1 \& C_2 \text{ are the concentration inner \& outer acid solutions.}$$

$$E_G = 0.0591 \cdot \log \frac{[C_2]}{[C_1]} \quad \dots\dots\dots 3 \text{ Or } E_G = 0.0591 \cdot \log \frac{[C_2]}{[C_1]}$$

$$E_G = -0.0591 \cdot \log [C_1] + 0.0591 \cdot \log [C_2] \quad \dots\dots\dots 4$$

Since the H^+ concentration inside the glass bulb is a constant. The first term on RHS of the above equation becomes a constant.

$$E_G = \text{Constant} + 0.0591 \cdot \log [C_2] \quad \dots\dots\dots 5$$

$$E_G = \text{Constant} + 0.0591 \cdot \log [H^+] \quad \dots\dots\dots 6 \quad \text{Since, } C_2 = [H^+]$$

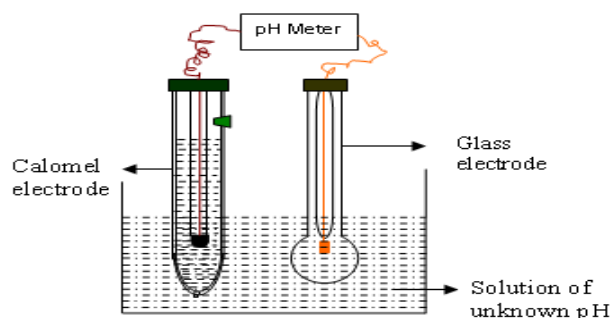
$$E_G = \text{Constant} - 0.0591 \cdot \text{pH} \quad \dots\dots\dots 7 \quad \text{Where pH} = -\log [H^+]$$

Advantages

1. This electrode can be used to determine pH in the range 0-9, with special type of glass even up to 12 can be calculated.
2. It can be used even in the case of strong oxidizing agents.
3. The equilibrium is reached quickly.
4. It is simple to operate, hence extensively used in various laboratories.

Determination of pH using Glass electrode

Principle: When a thin glass membrane is placed between two solutions of different pH values, a potential difference arises across the membrane. The potential difference varies as the pH of these solutions varies. In practice, pH of one of these solutions is kept constant and therefore the electrode potential depends on pH of the other solution i.e. Experimental solution.



Procedure: glass electrode is immersed in the solution; the pH is to be determined. It is combined with a reference electrode such as a calomel electrode through a salt bridge. The cell assembly is represented as,



The emf of the above cell, E_{cell} is measured using an electronic voltmeter with a pH meter.

The emf of the cell is given by

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \quad \dots\dots\dots 1$$

$$E_{\text{cell}} = E_{\text{glass}} - E_{\text{SCE}} \quad \dots\dots\dots 2$$

Since E_{SCE} is known EMF of the cell,

Substitute the value of E_{G} to equation (2)

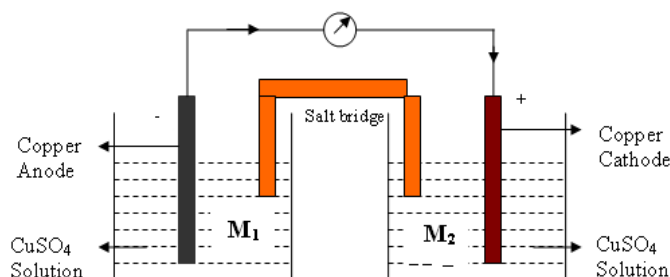
$$E_{\text{cell}} = \text{Constant} - 0.0591 \cdot \text{pH} - E_{\text{sce}} \quad \dots\dots\dots 3$$

$$\text{pH} = \frac{\text{Constant} - E_{\text{cell}} - E_{\text{sce}}}{0.0591} \quad \dots\dots\dots 4$$

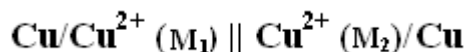
Concentration cell

“Concentration cell is an electrochemical cell in which the electrode material and the solution in both the electrodes are composed of the same substances but only the concentrations of the two solutions are different”.

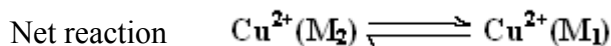
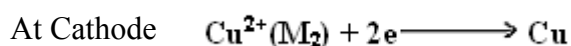
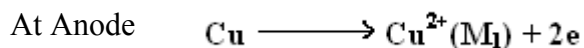
A typical example of Copper concentration cell is shown below.



It consists of two Copper electrodes are immersed in two different concentration of CuSO_4 solutions. These two electrodes are externally connected by metallic wire and internally by a salt bridge. The cell can be represented as,



By convention left hand electrode is the anode and right hand electrode is cathode.



The net cell reaction is merely the change in concentration as a result of current flow.

Emf of concentration cell:

We know that electrode potential depends upon the concentration of the electrolyte. By convention, the potential of the cell is

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} \quad \dots\dots\dots 1$$

$$E_{\text{cell}} = \frac{0.0591}{n} \cdot \log[M_2] - \frac{0.0591}{n} \cdot \log[M_1] \quad \dots\dots\dots 2$$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \frac{[M_2]}{[M_1]} \text{ at } 298\text{K} \quad \dots\dots\dots 3$$

From equation (3) following conclusions may be drawn.

1. When the two solutions are of the same concentrations, $\log \frac{[M_2]}{[M_1]} = 0$ and no electricity flows. Hence, $E_{\text{cell}} = 0$
2. When $M_2/M_1 > 1$ i.e $M_2 > M_1$, $\log M_2/M_1$ is positive & electrode potential is positive.
3. Higher the ratio of M_2/M_1 , higher is the value of cell potential.

Numerical problems.

1. Calculate the emf of concentration cell $\text{Cu} / \text{Cu}^{2+} (0.072\text{M}) \parallel \text{Cu} / \text{Cu}^{2+} (0.32\text{M})$.

$$\text{Solution:} \quad E_{\text{cell}} = 0.0591/n * \log [C_2 / C_1]$$

$$C_1 = 0.072\text{M}$$

$$C_2 = 0.32\text{M}$$

$$n = 2 \quad E_{\text{cell}} = 0.0591/2 * \log [0.32/0.072]$$

$$E_{\text{cell}} = 0.0295 * \log [4.444]$$

$$E_{\text{cell}} = 0.0295 * (0.6478)$$

$$E_{\text{cell}} = \mathbf{0.01914\text{ V}}$$

2. From the following cell $\text{Tin} / \text{Tin ion} (0.024\text{M}) \parallel \text{Tin ion} (0.064\text{M}) / \text{Tin}$
: $E_{\text{cell}} = 0.0126\text{ V}$. Calculate the valency of Tin.

Solution: $C_1 = 0.024\text{M}$, $C_2 = 0.064\text{M}$, $E_{\text{cell}} = 0.0126\text{ V}$, $n = ?$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \left[\frac{C_2}{C_1} \right]$$

$$n = \frac{0.0591}{E_{\text{cell}}} \log \left[\frac{C_2}{C_1} \right]$$

$$n = \frac{0.0591}{0.0126} \log \left[\frac{0.064}{0.024} \right]$$

$$n = 4.6904 \times \log (2.6666)$$

$$n = 4.6904 \times 0.4259$$

$$\mathbf{n = 1.997 = +2}$$

3. The emf cell $\text{Cd} / \text{CdSO}_4 (0.0093\text{M}) // \text{CdSO}_4 (\text{XM}) / \text{Cd}$ is 0.086 V . Find the value of X.

Solution: $C_1 = 0.0093\text{M}$ $C_2 = \text{XM} = ?$ $E_{\text{cell}} = 0.086\text{ V}$, $n = 2$

$$E_{\text{cell}} = \frac{0.0591}{n} \log \left[\frac{C_2}{C_1} \right]$$

$$0.086 = \frac{0.0591}{2} \log \left[\frac{X}{0.0093} \right]$$

$$0.086 = 0.0295 \times [\log X - \log 0.0093]$$

$$\log X - \log 0.0093 = 0.086 / 0.0295$$

$$\log X - (-2.0315) = 2.9152$$

$$\log X = 2.9152 - 2.0315$$

$$\log X = 0.8837$$

$$X = \text{Anti log } (0.8837)$$

$$\mathbf{X = 7.65\text{ M} = C_2}$$

1. Calculate the emf of concentration cell $\text{Zn} / \text{Zn}^{2+} (0.01\text{M}) // \text{Zn} / \text{Zn}^{2+} (0.1\text{ M})$.
2. Calculate the emf of concentration cell $\text{Ag} / \text{Ag}^+ (0.018\text{M}) // \text{Ag} / \text{Ag}^+ (1.2\text{ M})$.
3. Calculate the emf of concentration cell $\text{Fe} / \text{Fe}^{2+} (0.001\text{M}) // \text{Fe} / \text{Fe}^{2+} (0.01\text{ M})$.
4. Calculate the emf of concentration cell $\text{Li} / \text{Li}^+ (0.005\text{M}) // \text{Li} / \text{Li}^+ (0.05\text{ M})$.
5. Calculate the emf of concentration cell $\text{Ag} / \text{Ag}^+ (1\text{M}) // \text{Ag} / \text{Ag}^+ (1\text{ M})$.
6. From the following cell $\text{Zinc} / \text{Zinc ion} (0.01\text{M}) // \text{Zinc ion} (0.1\text{M}) / \text{Zin} : E_{\text{cell}} = 0.0295\text{ V}$. Calculate the valency of Zinc.
7. From the following cell $\text{Zinc} / \text{Zinc ion} (0.01\text{M}) // \text{Zinc ion} (0.1\text{M}) / \text{Zin} : E_{\text{cell}} = 0.0295\text{ V}$. Calculate the valency of Zinc.
8. The emf cell $\text{Cd} / \text{CdSO}_4 (\text{XM}) // \text{CdSO}_4 (0.025\text{M}) / \text{Cd}$ is 0.035 V . Find the value of X.

Sensors

A sensor is **a device that detects and responds to some type of input from the physical environment**. The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.

OR

A **sensor** is a device that produces an output signal for the purpose of **sensing** a physical phenomenon.

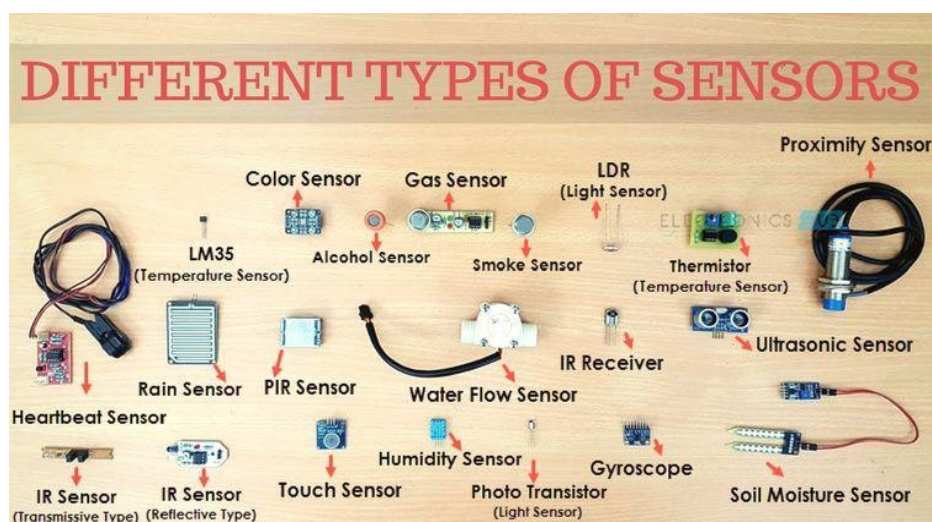
OR

A device that detects/measures a signal.

The word “sensor” comes from Latin (sensus: sense) and means feeler.

There are many different types of sensors, the main categories are;

- Position Sensors.
- Pressure Sensors.
- Temperature Sensors.
- Force Sensors.
- Vibration Sensors.
- Piezo Sensors.
- Fluid Property Sensors.
- Humidity Sensors.



Working Principle and Applications of Conductometric sensors

Conductometric detection is based on measurement of specific conductance of an analyte and is preferable because it can be applied for detection of both electroactive and electroinactive species.

Working Principle:

- The basic principle of conductometric detection involves a reaction that can change the ionic species concentration.
- This reaction leads to changes in electrical conductivity or current flow.
- In this method, two inert metal electrodes are used.

Application of conductometric sensor

- Conductometric sensors are used to determine water purity.
- It is used to check the levels of pollution present in different small water bodies like lakes, ponds or rivers.
- Can also be used to examine the salinity of seawater and the alkalinity of freshwater
- Used for the Detection of Food borne Pathogens and Toxins
- It is used for DNA Detection

Working principle and applications of Electrochemical sensors

Definition:

Electrochemical sensors are made up of three essential components: a receptor that binds the sample, the sample or analyte, and a transducer to convert the reaction into a measurable electrical signal. In the case of electrochemical sensors, the electrode acts as the transducer.

[Receptors are synthetic molecules having structural functional groups able to selectively interact with an analyte.

Transducer: Is an electronic device that converts energy from one form to another].

Working Principle:

Electrochemical sensors work by reacting with the gas of interest and producing an electrical signal proportional to the gas concentration. Consisting of two electrodes (a working

electrode and a counter electrode), the sensor operates by allowing charged molecules to pass through a thin layer of electrolyte.

Applications Electrochemical sensors

- Electrochemical sensors are considered attractive tools to determine important molecules or biomarkers that are used for the diagnosis of diseases and disorders.
- They are used for the monitoring of toxic levels of different substances in food quality and environmental control.
- The biosensor application areas of these sensors extend to medical and biomedical applications, process control, bioreactors, quality control, agriculture, bacterial and viral diagnosis, industrial wastewater control.

Working principle and applications of Thermometric sensors

Definition:

A temperature sensor is an electronic device that measures the temperature of its environment and converts the input data into electronic data to record, monitor, or signal temperature changes.

Working Principle:

Temperature sensors work by providing readings via electrical signals. Sensors are composed of two metals that generate an electrical voltage or resistance when a temperature change occurs by measuring the voltage across the diode terminals. When the voltage increases, the temperature also increases.

Temperature Sensor Applications

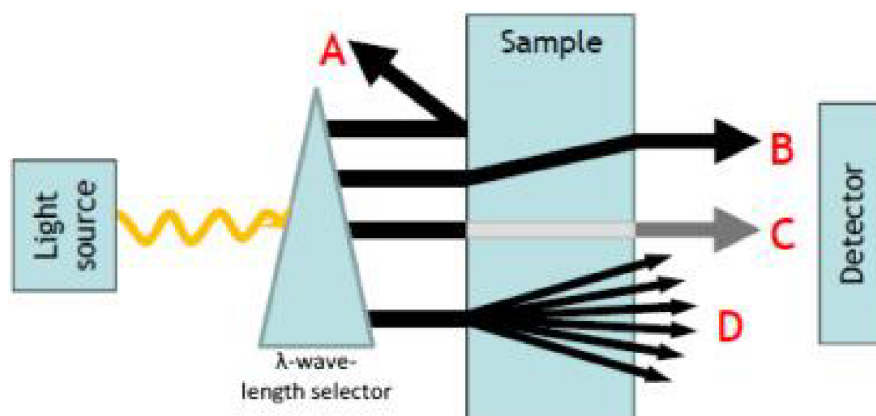
- The main application of these temperature sensors is in hazardous locations like nuclear power plants.
- Temperature sensors identify the heat of hydration in concrete structures in geothermal control. They can also be employed to control the migration of seepage or groundwater.
- They can detect rock temperatures in order to identify storage tanks of liquid gasses and the ground freezing process.

Working principle and applications of Optical sensors

Definition:

- Optical sensors have **the ability to detect light, typically at a specific range of the electromagnetic spectra (ultraviolet, visible, and infrared).**
- The sensor detects wavelength, frequency, or polarization of light and converts it into electric signal due to photoelectric effect.

Working Principle:



Optical sensors have the capacity to detect light usually at an exact electromagnetic spectra range like visible, infrared, and ultraviolet. This sensor simply detects the polarization of light, wavelength, or frequency and changes it into an electric signal because of the photoelectric effect. An optical sensor's detecting principle mainly depends on changes within the optical signal's characteristics. This sensor works mostly by using light as the medium, thus the distance of its detection is extremely long.

Applications of Optical sensors

- Optical sensors are integral parts of many common devices, including computers, copy machines (xerox) and light fixtures that turn on automatically in the dark.
- And some of the common applications include **alarm systems, synchros for photographic flashes and systems that can detect the presence of objects.**

Analytical Techniques

Introduction:

Analytical chemistry deals with the quantitative and qualitative characterization of a materials.it cover a wide variety of fields like environmental science, agricultural science,

biomedical and clinical chemistry, solid state research and electronics, oceanography, forensic research and space research. Analytical chemist is an integral part of the R and D sections of any industry.

Chemical analysis of a matter is of two types.

- 1) Qualitative analysis
- 2) Quantitative analysis

Principle and Instrumentation of Colorimetric sensors;

Principle: Colorimetric analysis depends upon the measurement of quantity of light absorbed by a colored solution. Quantitative analysis by colorimetric is based on Beer–Lambert

Law, which can be expressed by the relation

Lambert’s law: This law states that “When monochromatic light passes through a transparent medium, the rate of decrease in intensity with thickness of the medium is proportional to the intensity of the light”

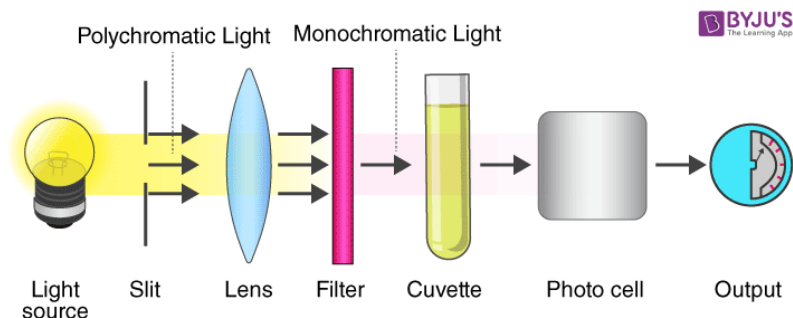
Beer’s law- This law states that “When monochromatic light passes through a transparent medium, the rate of decrease in intensity with concentration of the medium is proportional to the intensity of the light

Therefore absorbance of the colored solution changes with both concentration and thickness of the solution, if thickness kept constant, absorbance of the solution increases with increase in the concentration. On measuring the Absorbance of known and unknown concentration solutions with respect to blank solution. Concentration of unknown solution can be determined from the graph.

Instrumentation

The essential parts of colorimeter are

1. **Source** – Light
2. **Filter** - To filter undesired radiations , it allows radiations of a definite wave length Range to pass through it and reach the sample
3. **Cuvette-** To take sample
4. **Photocell** - To receive the transmitted light
5. **Recorder** – to record the absorbance in nm.



Application in the Estimation of Copper

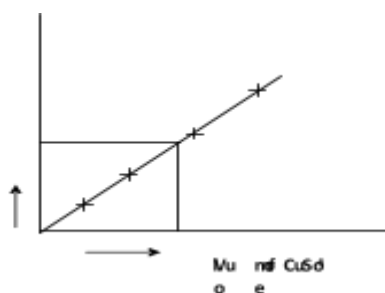
Quantitative Analysis: Colorimetric determination of copper in copper sulphate solution

From a burette, take 5, 10, 15, 20 cm³ of given copper sulphate solution into four separate 50 cm³ volumetric flask. From another burette, add 5 cm³ of ammonia to each one of them, to the unknown solution and blank solution flask. To the entire six flasks, add water up to the mark.

Mix well. Wait for 10 minutes. In the colorimeter set the wavelength of light to 620 nm. For the blank solution, set absorbance to zero. Then, measure the absorbance of other five solutions. Plot a graph of absorbance against volume of CuSO₄. Determine the volume of CuSO₄ in unknown solution from the graph and calculate the weight of Cu in the given solution.



Dark blue colour



Calculation:

1000 cm³ of stock solution contains 8.0 g of CuSO₄ · 5H₂O

249.54 g of CuSO₄ · 5H₂O ≡ 63.54 g of Cu

$$8.0 \text{ g of } \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \text{ contains } \frac{63.54 \times 8}{249.54} = 2.037 \text{ g of Cu per } 1000 \text{ cm}^3 \text{ of stock solution}$$

$$\therefore 1 \text{ cm}^3 \text{ of } \text{CuSO}_4 \cdot 5\text{H}_2\text{O} \equiv \frac{2.037}{1000} = 0.002037 \text{ g of Cu} = 2.037 \text{ mg of Cu}$$

Copper present in 'a' cm³ of test solution = 'a' cm³ X 2.037 mg = ... mg

Principle and instrumentation of Potentiometry

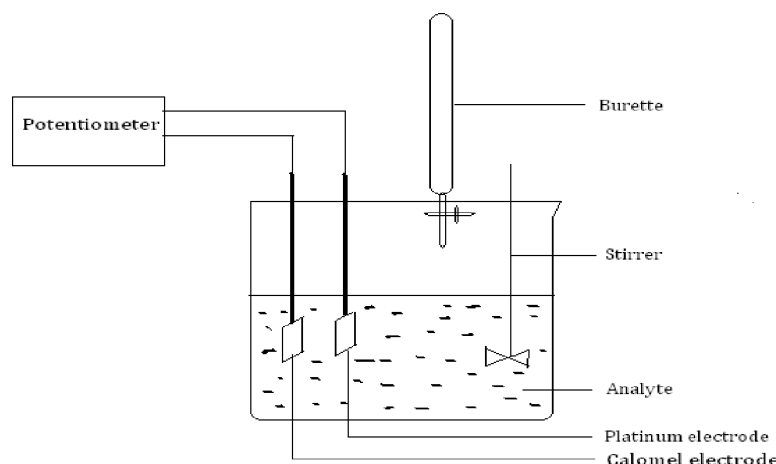
Principle: It is an electro analytical technique, in which the amount of substance in the solution is determined by measuring the emf between two electrodes that are dipped into the solution. When the metal M is immersed in the solution containing its own ions Mⁿ⁺ ions, the electrode potential is given by Nernst equation,

$$E = E_o + \frac{0.0591}{n} \log [\text{Mn}^+]$$

From this equation the potential of the electrode is directly proportional to the concentration of the ionic species present in the solution.

Instrumentation: A potentiometer consists of a indicator electrode (e.g.: Platinum) and an saturated reference electrode (E.g.: Calomel electrode), stirrer & potentiometer to measure the potential values.

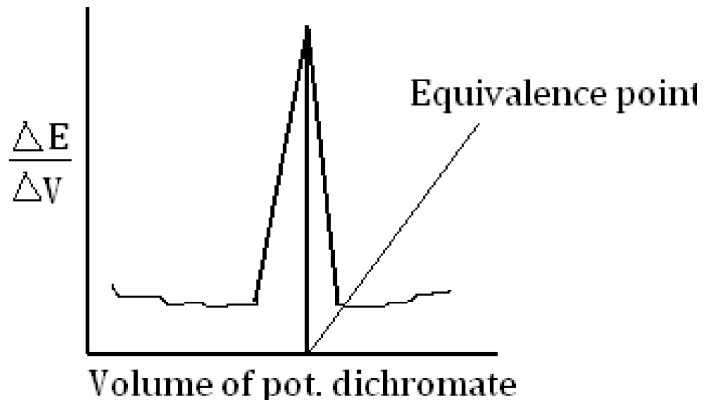
Emf of the solution can be measured by combining reference electrode with indicator electrode. The electrode which responds to the change in the concentration of the ion in the solution is called indicator electrode & reference electrode is one whose potential is constant. A known volume of the analyte is taken in beaker and its potential is measured. The titrant is added in increments of 0.5ml and potential is measured each time. Near the equivalence point there is a sharp increase in the potential. The end point is determined by plotting change in potential against volume of the titrant.



For example: Estimation of iron.

Procedure: Pipette out 25cm³ of FAS solution into a 50 cm³ beaker. Add one test tube full of dil H₂SO₄. Immerse Pt. & calomel electrodes into the solution, & connect the electrodes to a Potentiometer. Fill the burette with K₂Cr₂O₇ solution. Add K₂Cr₂O₇ solution from the burette with increment of 0.5cm³, stir well and measure the potential after each addition. Continue the titration till the potential indicates a rapid jump with a drop of titrant.

Plot the graph of $\Delta E / \Delta V$ v/s vol. of K₂Cr₂O₇.



Calculation:

$$\text{Normality of FAS} = \frac{(N \times V) K_2Cr_2O_7}{\text{Volume of FAS}} = \dots \dots \dots$$

$$\text{Weight of FAS per liter} = \text{Normality} \times \text{Equivalent weight of FAS (392)} = \dots \dots \dots (b).$$

$$\text{Weight of FAS per } 25\text{cm}^3 = \frac{b \times 25}{1000} = \dots \dots \dots \text{g/cm}^3$$

Principle and instrumentation of Conductometry

Conductance is ease with which current flows through the solution. It is reciprocal of resistance. $C = 1/R = \Omega^{-1}$ or mho or siemen

Principle:

The Conductance of the solution is explained by considering ohm's law.

According to ohm's law the current flowing through the conductor is directly proportional to voltage and inversely proportional to the resistance.

$$I = \text{or } E = I R$$

The resistance of the any conductor is directly proportional to the length of inversely proportional to the area of cross section of the conductor

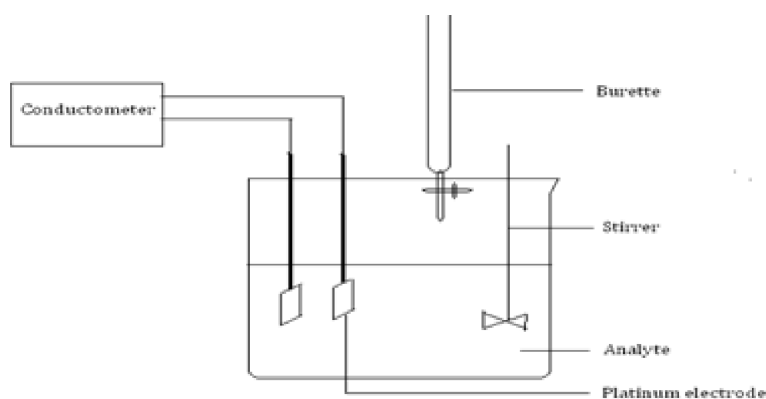
Therefore $R = S (l/a)$ where S is specific resistance

Therefore $C = 1/R = 1/S (a/l)$, $K (a/l)$, K = specific conductance

It is defined as the conductance of the solution present between two parallel electrodes of 1cm^2 area of cross section and 1 cm apart.

The conductance of the solution is depends on mobility of the ion and number of the ion.

Instrumentation: It consists of two platinum electrodes each of unit area of cross section placed at unit distance apart. The electrodes are dipped in the electrolytic solution taken in a beaker. It is connected to a conductometer to measure the conductance. The conductance is measured after the addition of the titrant at intervals of 0.5 ml.



Application of Conductometric titration

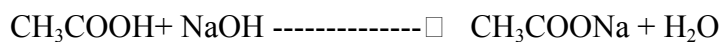
It is used in the estimation of acids and bases present in the sample solutions. The possible combinations and conductance responses are explained below,

- ☐ Strong acid v/s strong base
- ☒ **Weak acid v/s Strong base**
- ☐ Mixture of strong acid and weak acid v/s strong base

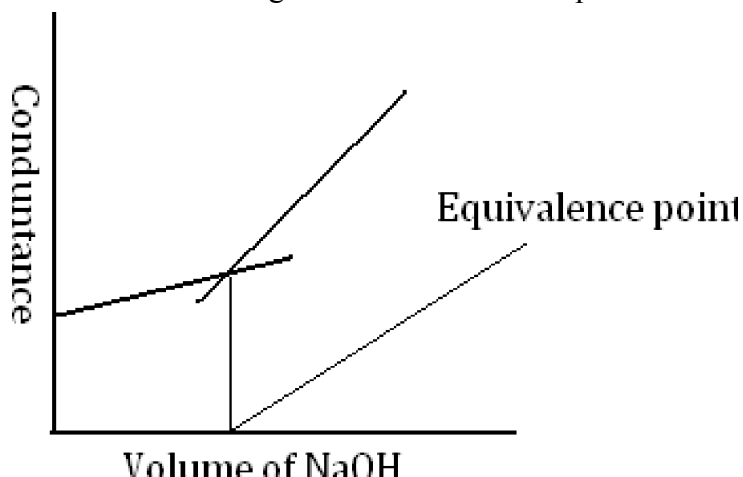
Weak acid v/s Strong base:

Consider the titration of acetic acid against NaOH.

The conductance of the acid will be initially low since acetic acid is a weak electrolyte. When NaOH is added to the acid, the salt formed is highly ionized and the conductance increases. On complete neutralization of the acid, further addition of base leads to an increase in the number of mobile OH⁻ ions. Hence the conductance increases sharply.



A plot of conductance against the volume of base added is shown in the figure. The point of intersection of two curves gives the neutralization point.

**Calculation :****Estimation of CH₃COOH**

Volume of NaOH required to neutralize **CH₃COOH** = V cm³

Normality of **CH₃COOH** = (N X V)NaOH / volume of acetic acid

Therefore, the weight of CH₃COOH /dm³ = Normality of CH₃COOH X Equivalent weight of CH₃COOH (60.0)

= g/dm³

Application:

1. Estimation of concentration of the ionic species present in the given solution.
2. Coloured solution can also be titrated
3. Acid-base titration can also be done in this method.
4. In this method Oxidation-reduction titrations can also be carried out.
5. Precipitation reactions can also be carried out potentiometrically.

MOST REPEATED QUESTIONS IN VTU EXAM

1. What is reference electrode? Describe the construction & working of Calomel electrode (7 Marks)
2. What are Concentration cells? Explain the construction & working of electrolyte concentration cell with a suitable example. (7 Marks)
3. What are ion selective electrodes? Describe the construction & working of Glass electrode. (7 Marks)
4. Explain the determination of pH using glass electrode (6 Marks)
5. **Problems on Concentration cell** (6 Marks)
6. What are optical sensors? Explain the Principle and Instrumentation of Colorimetric sensors. (7Marks)
7. What are Potentiometric sensors? Explain the working Principle, Instrumentation and Applications of Potentiometric sensors (Estimation of Iron) (7Marks)
8. What are Conductometric sensors? Explain the working Principle, Instrumentation and Applications of Conductometric sensors (Estimation of Weak acid) (7Marks)
9. What are Electrochemical sensors? Explain the working Principle and Applications of Electrochemical sensors (6Marks)
